

4.5.1 Photons

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the particulate nature (photon model) of electromagnetic radiation	
(b) photon as a quantum of energy of electromagnetic radiation	
(c) energy of a photon; $E = hf$ and $E = \frac{hc}{\lambda}$	
(d) the electronvolt (eV) as a unit of energy	
(e) (i) using LEDs and the equation $eV = \frac{hc}{\lambda}$ to estimate the value of Planck constant h (ii) Determine the Planck constant using different coloured LEDs.	No knowledge of semiconductor theory is required. PAG6